Identification and Structural Assessment of Alkaline-Earth Metal Complexes with Flavonols by FAB Mass Spectrometry

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Abstract - Complexation of a series of flavonols with Mg^{2+}, Ca^{2+}, and Ba^{2+} ions was studied by FAB mass spectrometry. The mass spectra of flavonol solutions containing magnesium ions revealed formation of 1 : 2 complexes MgL_2, undetectable by spectrophotometric and spectrofluorimetric methods. Experiments with a crown-substituted flavonol revealed formation of not unly previously known chelate complexes MgL and Mg_2L, but also of the 1 : 2 complex MgL_2 and crown complex MgHL. The barium ions form with the crown-substituted flavonol two types of "sandwich" complexes: $Ba(HL)_2$ and $Ba_2(HL)_2$. With the Ca^{2+} ions, such complexes are not formed, and only the fragmentation product of the complex $Ca(HL)_2$ was detected.